

Course:	ENGR 5785G: Real-time Data Analytics for IoT
Topic:	Feature Extraction & Feature Selection
Due:	As in Canvas
Weight:	5% of course grade
Format:	Jupyter Notebook (.ipynb) + Comment cells
Groups:	Individual submission only

Assignment Overview

In the in-class activity you applied feature engineering to the Titanic dataset with guidance. In this assignment you work independently on a new dataset, the Medical Cost Personal Dataset (insurance.csv), to predict whether a patient's medical charges are above the median (binary classification). You will demonstrate mastery of feature extraction, feature selection, model comparison, and written analysis.

Dataset Description

Dataset: **Medical Cost Personal Dataset** (also known as *insurance.csv*)

Source: Kaggle — available at kaggle.com/datasets/mirichoi0218/insurance

Column	Type	Description
age	int	Patient age in years (18–64)
sex	string	male / female
bmi	float	Body Mass Index — weight(kg)/height(m) ²
children	int	Number of dependents covered
smoker	string	yes / no — tobacco smoker
region	string	US region: northeast/southeast/southwest/northwest
charges	float	Annual medical cost billed — THIS is your source for the TARGET

Target variable (create it): `high_charges` = 1 if `charges > median(charges)`, else 0

Assignment Parts

Part A: Exploratory Data Analysis (10 points)

Before engineering any features, understand the raw data. Complete all of the following in clearly labelled notebook cells.

A1. Data Loading & Inspection (5 pts)

- Load the dataset and display the first 10 rows, dtypes, and `.describe()` summary.
- Check for missing values — report the count and % for each column.

- Report the class balance of your target variable (how many 0s vs 1s?).

A2. Univariate & Bivariate Analysis (5 pts)

- Plot the distribution of charges (histogram + KDE). Note whether it is skewed and explain the implication for modelling.
- Create a box plot of BMI grouped by smoker status. What pattern do you observe?
- Plot a grouped bar chart of mean charges by region and sex.
- Compute and display the correlation matrix for all numeric variables. Which pair has the highest correlation?
- Written answer (3–5 sentences): *Describe two findings from your EDA that suggest a useful feature to engineer.*

Part B: Feature Extraction (30 points)

Create at least 6 new features. For each feature, provide: (1) the code, (2) a brief justification, and (3) one plot or summary statistic showing its relationship with the target.

B1. Required Extractions (20 pts — 4 pts each)

1. **BMI Category:** create a categorical variable: Underweight (<18.5), Normal ($18.5\text{--}24.9$), Overweight ($25\text{--}29.9$), Obese (≥ 30). Encode as ordinal integers.
2. **Smoker $\times$ BMI interaction:** create a numeric feature: `smoker_bmi = smoker_encoded  $\times$  bmi`. Justify why this interaction may be predictive.
3. **Age Group:** bin age into: Young Adult ($18\text{--}35$), Middle-Aged ($36\text{--}50$), Senior ($51+$). Encode as integers.
4. **Family Risk Score:** combine children and age to create a composite risk metric of your own design. Justify your formula.
5. **Log Charges:** apply $\log_{10}$ transformation to charges before computing the target. Plot the before/after distribution. Explain the statistical benefit.

B2. Your Own Feature (10 pts)

Design and implement 1 additional feature not listed above. It must be:

- **Justified:** explain the domain logic or data insight that motivated it
- **Visualised:** include a plot showing its relationship with `high_charges`
- **Evaluated:** run a chi-squared test or point-biserial correlation to quantify its relevance

Part C: Feature Selection (30 points)

Apply three different selection methods and document your findings in a comparison table.

C1. Filter Method — Correlation & Chi-Squared (10 pts)

- Compute Pearson correlation of all **numeric features** with `high_charges`. Plot a bar chart of correlations ranked by absolute value.

- Run **SelectKBest(chi2, k=6)** on all non-negative features. Report the selected feature names and their χ^2 scores.
- Drop any feature with $|r| > 0.85$ with another feature (multicollinearity). Justify each drop.

C2. Embedded Method — Random Forest Importance (10 pts)

- Train a **RandomForestClassifier(n_estimators=100, random_state=42)** on all features. Plot feature importances as a horizontal bar chart.
- List the top 5 features by importance. Do they match the filter method results?
- Train a second RF using **only the bottom 5 features** by importance. Compare its cross-validated accuracy to the full-feature RF. What does this tell you?

C3. Wrapper Method — Recursive Feature Elimination (10 pts)

- Use **RFE(LogisticRegression(max_iter=1000), n_features_to_select=6)**. Report which features were selected.
- Produce a comparison table showing which features were chosen by each of the three methods.

Required Table: Feature Selection Summary

Create a table with features as rows and columns: [Correlation Rank | Chi2 Selected | RF Importance Rank | RFE Selected | Your Verdict]. Mark each cell ✓ or X. Add a 'Your Verdict' column with your final feature set choice and a brief justification.

Part D: Model Comparison (20 points)

Train and evaluate Logistic Regression models under three experimental conditions using 5-fold cross-validation.

D1. Three Experiments (10 pts)

Experiment	Features Used	Report
Baseline	Raw cleaned features only (age, sex, bmi, children, smoker, region encoded)	Accuracy, F1, AUC-ROC
Extracted	All original + all 6 engineered features	Accuracy, F1, AUC-ROC
Selected	Your chosen top-6 features from Part C	Accuracy, F1, AUC-ROC, confusion matrix

D2. Bonus Challenge (bonus pts)

- Replace Logistic Regression with **Gradient Boosting (XGBoost or sklearn GradientBoostingClassifier)**. Does feature engineering still help? Explain why or why not (3–5 sentences).

Part E: Written Analysis & Reflection (10 points)

Write a clear, structured analysis of your results in comment cells.

E1. Results Summary (4 pts)

- Summarize your three experiments in a results table (Experiment | Accuracy | F1 | AUC-ROC).
- Identify which single engineered feature produced the biggest gain. Justify your reasoning with evidence from your feature importance or correlation analysis.

E2. Critical Analysis (4 pts)

- Explain a case where feature extraction HURT performance. Why might adding too many engineered features be harmful?
- Describe one feature you tried that did **not** improve the model. What does that tell you about the underlying data?

E3. Reflection (2 pts)

- If you were deploying this model in a real insurance context, which features would you be cautious about using, and why? (Consider fairness and legal constraints.)

Grading Rubric

Section	Criteria	Points
Part A: EDA	Completeness of analysis, quality of plots, written insight	10
Part B1: Required Extractions	Correct implementation, justification, supporting visualization	20
Part B2: Original Feature	Creativity, domain logic, statistical validation	10
Part C1: Filter Selection	Correct correlation/chi2 analysis, multicollinearity handled	10
Part C2: RF Importances	Correct RF training, chart, bottom-5 comparison insight	10
Part C3: RFE + Summary Table	Correct RFE implementation, complete comparison table	10
Part D: Model Comparison	3 experiments run, metrics reported, correct cross-validation	20
Part E: Written Analysis [comments]	Clarity, evidence-backed arguments, fairness reflection	10
Notebook Quality	Clean code, markdown explanations, runs without errors	—
TOTAL		100 pts

Submission Requirements

- **File 1:** `LastName_FirstName_FeatureEng.ipynb` — Jupyter notebook, runs top-to-bottom without errors
- **Data:** Do NOT submit the dataset file — it will be sourced from Kaggle
- **Libraries:** pandas, numpy, scikit-learn, matplotlib, seaborn — no other packages without instructor approval
- **Late policy:** –10% per day, no submissions accepted after 5 days

Academic Integrity Reminder

You may use official documentation and lecture notes. Using generative AI tools to write your analysis or code is not permitted. Feature engineering requires you to think about the data — let the data guide your decisions, not an auto-complete tool. All submissions are checked for similarity.

Starter Code Template

Copy this into the first cell of your notebook to get started:

```
# — Imports —————
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.preprocessing import LabelEncoder, StandardScaler
from sklearn.ensemble import RandomForestClassifier
from sklearn.linear_model import LogisticRegression
from sklearn.feature_selection import SelectKBest, chi2, RFE
from sklearn.model_selection import cross_val_score, StratifiedKFold
from sklearn.metrics import classification_report, roc_auc_score

# — Load data —————
df = pd.read_csv('insurance.csv')

# — Create binary target —————
df['high_charges'] = (df['charges'] > df['charges'].median()).astype(int)
print('Class balance:', df['high_charges'].value_counts())

# — Encode categorical columns —————
le = LabelEncoder()
df['sex'] = le.fit_transform(df['sex']) # female=0, male=1
df['smoker'] = le.fit_transform(df['smoker']) # no=0, yes=1
df['region'] = le.fit_transform(df['region']) # 4 classes → 0..3

# — YOUR CODE STARTS HERE —————
# Part A: EDA ...
# Part B: Feature Extraction ...
# Part C: Feature Selection ...
# Part D: Model Comparison ...
```